

Annexure - A

USAGE OF ARTIFICIAL INTELLIGENCE (AI) MODELS

1. Artificial Intelligence (AI) refers to the simulation of human Intelligence in machines that are programmed to think, learn and make decisions. It encompasses a wide range of technologies, including machine learning, natural language processing, computer vision and robotics. AI systems can analyse vast amounts of data, recognize patterns and perform multifarious tasks. However, it also raises data privacy and ethical concerns.

2. Commonly used AI models include conversational models, contextual search, image generation, speech recognition, video generation, gaming, simulation, semi-autonomous, content generation, data analysis and forecasting models. However, they have associated risks of exposing personal data, biased opinions, lack of transparency, surveillance, technical risks and data storage in foreign country. Commonly used AI models are as under: -

Sr. No.	Model	Application Area
a)	Open AI's GPT-4	Chatting, content generation, contextual search and creative writing
b)	Google's BERT	contextual search, Natural Language Processing (NLP)
c)	Open AI's DALLE	Image Generation, Creative Design
d)	DeepSeek R1	Chatting, content generation and reasoning

3. **Working of Online/ Cloud Hosted AI Models:** The end user writes a query which is taken as a prompt by the model. This query is sent to the AI server, where it is processed, biased as per its own filters and response is sent to the user. In the backend, data is processed in the following manner without explicit knowledge of the user: -

- AI model may keep a backup of previous queries, expected reply formats, preferences and other customisations. This information is used to modulate future response. Though this feature was designed to improve user experience, however if breached by hackers or required by government agencies, this data can be used to profile an individual, gather intelligence and make predictions for future decisions.
- The AI servers of the popular models are usually located outside India, thereby having limited or no data autonomy.
- AI model responses are modulated and biased on national policies and goals of their origin countries.

4. **Risks:** Common AI platforms have inherent risks of data collection, user input storage, keystroke pattern recording, undefined data retention periods, data sharing with third parties, access by respective government, international and jurisdictional issues, lack of granular consent mechanism, potential data breaches, unilateral policy changes, automatic data collection like IP address, device ID, keystroke patterns.

5. **Sample Scenarios of Risks to Military Personnel:** Several AI platforms have surfaced in open domain especially in the last 3-4 years. AI platforms like Deep Seek which have direct linkage with China collect considerable personal as well as network information pertaining to users who access the application. Sample scenarios and risks are associated with their usage are highlighted in Annexure I. These need to be factored in when making decisions to use such applications/ models for personal/ organizational purpose.

6. **Dos & Don'ts:** Considering the threats, risks and vulnerabilities in use of AI models, it is difficult to remain immune to its misuse. However, if in case it is considered inevitable, following measures may be undertaken to achieve some level of security. List of Dos & Don'ts for end user are highlighted at Annexure II.

7. **Offline Deployment of AI Models:** Offline deployments of certain AI models may be used, in case of specific organizational requirements. Precautions to be taken during offline deployment include use of open-source models, deployment in air gapped environment, use of encryption to secure data, use of virtual machines or containers and making AI model specific block rules in network firewall.

8. End users should be sensitized that AI models have very limited security features to protect data privacy. Hence, AI models are not to be used for official work and great caution to be exercised when used for personal requirements. In case such technology is required to be used for organizational requirements, then deployment should be used in standalone/ offline mode, post obtaining approval of competent authority.

9. For information and further dissemination please

SAMPLE SCENARIOS OF RISK TO DEFENCE PERSONNEL

Scenario 1

1. **Location & Deployment Tracking:** The concerned wings/ section access Deep Seek from a secure base to seek assistance with document translations or technical queries. Since Deep Seek logs IP addresses, device identifiers and behavioural patterns, adversaries monitoring this data could track frequent logins from specific locations. Impact is as under: -

- a) Strategic locations/ sections could be mapped and targeted for cyber-attacks, surveillance or even kinetic attacks.

Scenario 2

2. **Data Breach Leading to Social Engineering Attacks:** Anti-national elements hack Deep Seek's servers and steal chat histories, device data and user interactions. They analyse stored conversations of military personnel discussing work related matters. Impact is as under:

- a) Attackers use extracted data to craft highly convincing phishing emails or spear-phishing attacks on military personnel.
- b) They impersonate trusted contacts using leaked chat histories to gain access to classified networks or extract further intelligence.
- c) Psychological profiling of individuals could be used for blackmail, coercion or insider recruitment by adversarial intelligence agencies.

Scenario 3

3. **Compromised AI Outputs Leading to Disinformation:** If anti national hackers infiltrate Deep Seek's AI model, they could manipulate its responses. Military personnel relying on Deep Seek for open-source intelligence, translations or quick research could receive altered or misleading information. Impact is as under: -

- a) Misleading translations could distort intelligence reports, leading to miscommunication between allies.
- b) Disinformation campaigns could be executed through manipulated AI responses, causing operational disruptions.
- c) Adversaries could plant false narratives that influence strategic decisions.

Scenario 4

4. **Targeted Leaks & Intelligence Gathering:** A military strategist uses Deep Seek for document analysis and AI-assisted planning. If Deep Seek is legally obligated or compromised to share user interactions, adversaries could monitor key operational terms and strategies. Impact is as under: -

- a) Enemy forces gain pre-emptive knowledge of upcoming military exercises or defence strategies.
- b) Sensitive information about alliances, supply chains or vulnerabilities could be exploited in cyber or physical warfare.
- c) Deep fake content could be created based on extracted information, affecting diplomacy & defence strategies.

IMPORTANT Dos AND Don'ts FOR USE OF AI MODELS

DO's

1. Use of on-premises offline solutions
2. Prefer open source, audited AI models
3. Ensure AI model complies with Data Privacy laws -DPDP Act, GDPR etc
4. Use only for non-sensitive tasks
5. Apply Data Masking & Sanitisation before submitting to AI model
6. Monitor for Data Exfiltration Risks
7. Beware of AI Generated Phishing & Deepfakes
8. Verify AI Output for Bias & Misinformation
9. Restrict API Access to API models
10. Modify Firewall blacklist to block/ grant selective access using IAM.
11. Prepare a local policy for use of AI models based on ICSP guidelines.
12. Educate users on AI risks, cybersecurity and misinformation.
13. Remove Personal Identifiable Information & Log out of account
14. Anonymise prompts/ data inputs
15. Use disposable accounts to hide identity.
16. Clear AI chat histories & cache.
17. Disable browser autofill for AI sites.
18. Use Explainable AI (XAI) Models.
19. Check Model Hashes & Integrity before deployment and during updates.
20. Understanding that the AI responses can be biased based on their country of origin or intent
21. Enforce Zero Trust Architecture (ZTA) before deploying AI models.
22. Opt out of all optional features.
23. Minimise data sharing.
24. Understand third party policies.
25. Use AI models without an account or a fictitious account so that your original identity is preserved.
26. Use in a sandboxed environment, whenever possible.
27. Use a separate browser without cookies, history or stored accounts.
28. Use Data Loss Prevention (DLP) solutions to detect and block transmission of data.

DON'Ts

1. Don't share official information or other information which can reveal official designation, work, office or similar official details.
2. Never input Classified or Sensitive Information.
3. Disable data logging & telemetry (AI model sending data back to developer).
4. Don't connect AI systems to Operational NW & use these in Controlled Environments only. Enable RBAC (Role Based Access Control) and MFA (Multi Factor Authentication) during AI deployment.
5. Don't trust AI generated content without independent verification.
6. Don't use AI tools from untrusted, foreign or unknown vendors.